T3000 API & Protocol

LCD Screen Designer · built Nov 13, 2026

LCD Screen Designer

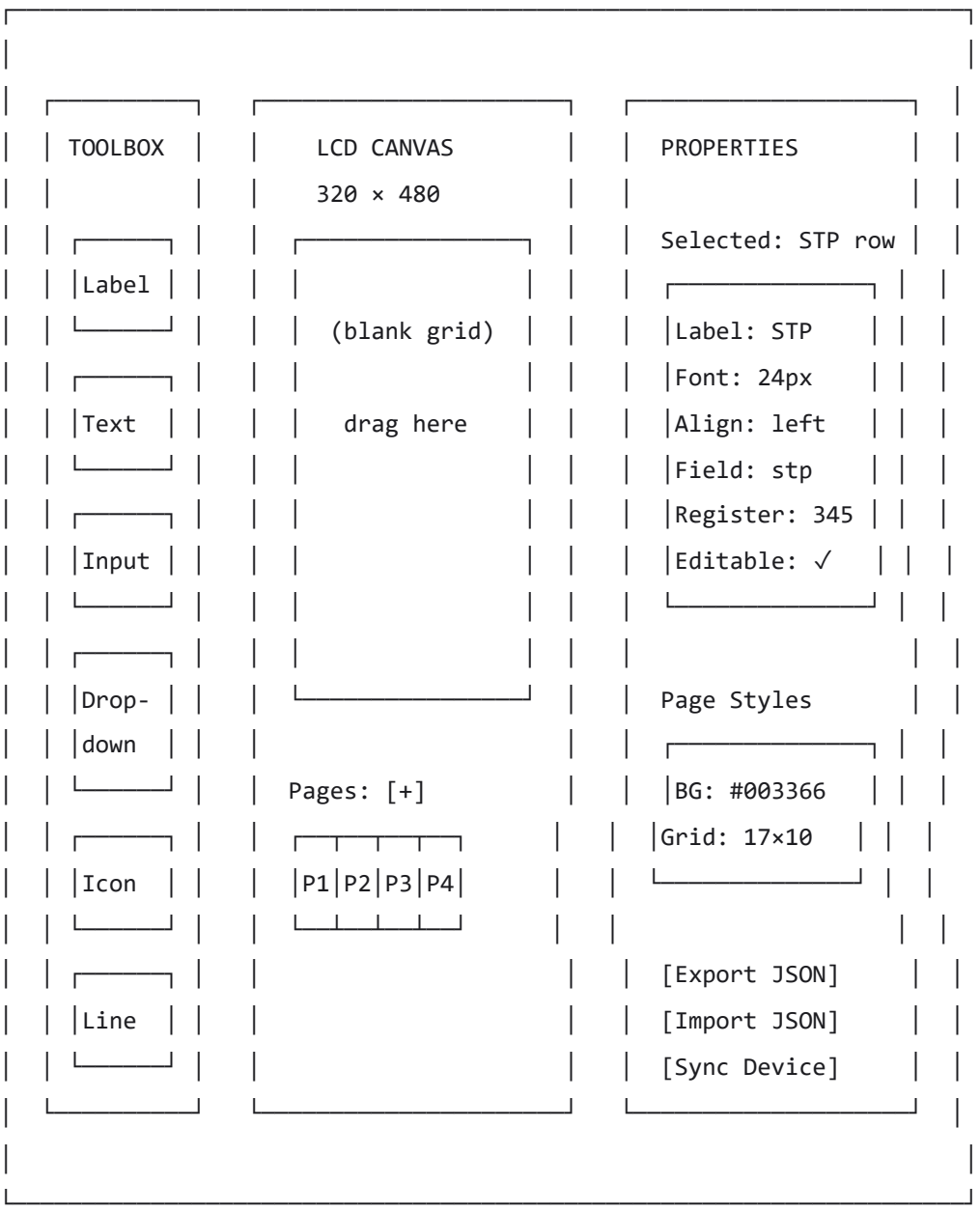
Visual drag-and-drop editor for creating LCD screen layouts from a blank canvas.

Core Concept

The designer starts with an **empty 320×480 canvas** — a blank LCD screen. A **Widget Toolbox** panel provides categorized widget types that the designer drags onto the canvas. Widgets snap to the 17×10 character grid. The designer adjusts each widget's position, text, font size, alignment, and data bindings. Multiple pages can be created. The final output is exported as `screenDefinition.json` and synced to the device.

This is the primary design tool — not a secondary "edit" mode on top of existing pages. The hardware screens shown in photos (Main, Setpoint, Inside, Comm Settings) are simply **example layouts** that were created using this designer.

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3-Panel Layout

Panel	Position	Purpose
Widget Toolbox	Left	Categorized list of draggable widget types
LCD Canvas	Center	The 320×480 design surface with grid overlay
Properties Panel	Right	Edit properties of selected widget + page styles

Widget Toolbox

The toolbox provides all available widget types organized by category:

Display Widgets

Widget	Icon	Description	Drag Behavior
Label	`Aa`	Static text label	Drops at grid position, default "Label"
Text	`T`	Data-bound text (shows register value)	Drops at grid position, pick data field
Large Text	`T+`	Big number display (temperature, etc.)	Spans multiple rows, centered
Header	`H`	Page title text	Drops at top, centered, larger font

Input Widgets

Widget	Icon	Description	Drag Behavior
Input Row	`[=]`	Label + editable value pair	Full-width row, label left + value right
Dropdown Row	`[▼]`	Label + value with dropdown options	Full-width row, configure options list
Edit Value	`[#]`	Standalone editable number in a box	Centered, large font, with +/-

Decorative Widgets

Widget	Icon	Description	Drag Behavior
Icon	☆	Single icon (house, wifi, fan, etc.)	Pick from icon palette
Icon Bar	☆☆☆	Row of navigation icons	Full-width, pick icons
Divider	–	Horizontal separator line	Full-width at grid row
WiFi Icon	≋	WiFi signal indicator	Fixed top-right corner

Footer Widgets

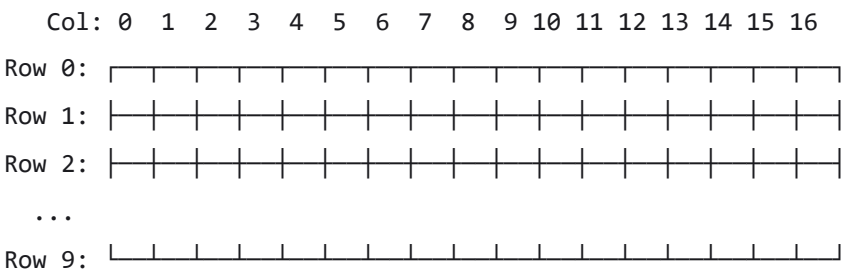
Widget	Icon	Description	Drag Behavior
Hint Text	`...`	Button hint text ("+ Edit -")	Bottom area
Nav Text	◄►	Navigation hint ("< Back Next >")	Bottom area

Grid Snapping & Alignment

Character Grid

The canvas is divided into a **17 column × 10 row** character grid:

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- **Cell size:** ~18.8px wide × 48px tall
- All widgets snap to grid cell boundaries when placed
- Widgets can span multiple cells (colSpan, rowSpan)

Snap Behavior

Action	Behavior
Drag from toolbox	Widget follows cursor, shows ghost position snapped to nearest grid cell
Drop on canvas	Widget snaps to the grid cell under the cursor
Drag to reposition	Widget moves between grid cells, snapping on release
Resize	Widgets resize in grid-cell increments (colSpan/rowSpan)

Alignment Aids

Feature	Description
Grid overlay	Visible grid lines showing all 17×10 cells
Snap indicators	Blue highlight showing which cell the widget will snap to
Alignment guides	Lines appear when widget aligns with edges of other widgets
Center markers	Guides for horizontal/vertical centering
Overlap warning	Red highlight if dropping on an occupied cell

Properties Panel

When a widget is selected on the canvas, the Properties Panel shows its editable properties:

Common Properties (all widgets)

Property	Control	Description
Position	Row / Col inputs	Grid row and column (with nudge arrows)
Span	ColSpan / RowSpan	How many cells the widget covers
Font Size	Dropdown	12px, 14px, 18px, 24px, 36px, 48px, 72px
Font Weight	Toggle	Normal / Bold
Text Align	Buttons	Left / Center / Right
Color	Color picker	Text color
Delete	Button	Remove this widget

Label / Text Properties

Property	Control	Description
Text	Text input	The static text to display
Field	Dropdown	Data field to bind (for dynamic text)
Suffix	Text input	Unit text after value ("°C", "%")

Input Row / Dropdown Properties

Property	Control	Description
Label	Text input	Left-side label text
Field	Dropdown	Data field to bind
Register	Number input	Modbus register number
Editable	Checkbox	Can the user change this value at runtime
Options	Tag input	List of allowed values (for dropdown)
Min / Max	Number inputs	Range for numeric values
Step	Number input	Increment step (e.g., 0.5)

Icon Properties

Property	Control	Description
Icon	Icon picker palette	Choose from available icons
Size	Dropdown	24px, 28px, 32px, 48px
Opacity	Slider	0.0 to 1.0

Page-Level Properties (shown when no widget selected)

Property	Control	Description
Page ID	Text input	Unique identifier
Page Label	Text input	Display name
Background	Color picker	Page background color
Highlight	Color picker	Focused widget color
Default Font	Dropdown	Base font for all widgets

Design Flow

Creating a New Screen from Scratch

```
1. Click [+ New Page] → blank 320×480 canvas appears
2. Grid overlay visible (17×10 cells)
3. Drag "Header" from toolbox → drop at row 1 → type "Communication\nSettings"
4. Drag "Input Row" → drop at row 4 → set label="Modbus", field="modbus", register=6
5. Drag "Input Row" → drop at row 5 → set label="Baud", field="baud", options=[9600,1152
6. Drag "Input Row" → drop at row 6 → set label="Mode", field="mode", options=["MODBUS",
7. Drag "Hint Text" → drop at row 8 → type "+  Edit  -"
8. Drag "Nav Text" → drop at row 9 → type "< Back   Next >"
9. Adjust fonts, alignment, colors in Properties Panel
10. Click [Export JSON] → downloads screenDefinition.json
11. Click [Sync Device] → pushes to hardware
```

Recreating the Main Display from Photos

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```
1. [+ New Page] → blank canvas
2. Drag "WiFi Icon" → auto-places at row 0, col 16
3. Drag "Large Text" → row 1, bind field="temp", suffix="°C", fontSize=72px
4. Drag "Divider" → row 3
5. Drag "Input Row" → row 3, label="STP", field="stp", editable=true
6. Drag "Input Row" → row 4, label="FAN", field="fan", options=["AUTO",-2,-1,0,1,2,3]
7. Drag "Input Row" → row 5, label="SYS", field="sys", options=["AUTO",0,1,2]
8. Drag "Text" → row 6, bind field="ip", fontSize=18px
9. Drag "Icon Bar" → row 9, pick icons: moon, house, snowflake, fan
10. Done — preview matches the hardware photo
```

Editing an Existing Layout

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```
1. Click [Import JSON] → load existing screenDefinition.json
2. All pages appear with widgets rendered on canvas
3. Click any widget → selected (blue outline, handles appear)
4. Drag to reposition → snaps to new grid cell
5. Edit properties in right panel
6. Add new widgets from toolbox
7. Delete unwanted widgets (select → Delete or ×)
8. Export updated JSON
```

Page Management

Page Tabs

Below the canvas, page tabs show all pages:

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```
┌──────┐ ┌──────────┐ ┌────────┐ ┌──────────────┐ ┌───┐
│ Main │ │ Setpoint │ │ Inside │ │ Comm Settings │ │ + │
└──────┘ └──────────┘ └────────┘ └──────────────┘ └───┘
  ↑ active
```

Action	Behavior
Click tab	Switch to that page
[+] button	Add new blank page
Right-click tab	Rename, duplicate, delete, reorder
Drag tab	Reorder pages (changes navigation order)

Templates (Optional)

For convenience, the designer can start from templates instead of a blank canvas:

Template	Pre-built Widgets
Blank	Empty canvas (default)
Main Display	Large temp + rows + icon bar
Settings Page	Header + input rows + footer
Info Display	Icons + read-only text

Templates are just pre-made JSON that gets loaded — the designer can modify everything.

Keyboard Shortcuts

Key	Action
Delete / Backspace	Delete selected widget
Ctrl+C	Copy selected widget
Ctrl+V	Paste widget
Ctrl+Z	Undo
Ctrl+Shift+Z	Redo
Ctrl+S	Export JSON
Arrow keys	Nudge selected widget by 1 grid cell
Ctrl+A	Select all widgets
Escape	Deselect

Data Model

TYPESCRIPT

```typescript
interface DesignerState {
  definition: ScreenDefinition;     // The full multi-page JSON
  activePageId: string;             // Currently viewed page
  selectedWidgetId: string | null;  // Currently selected widget for editing
  isDirty: boolean;                 // Unsaved changes
  clipboard: Widget | null;         // Copy/paste buffer
  history: ScreenDefinition[];      // Undo stack
  historyIndex: number;             // Current position in undo stack
  gridVisible: boolean;             // Show/hide grid overlay
  snapEnabled: boolean;             // Grid snap on/off
}
```

View Mode vs Design Mode

Aspect	Design Mode	View Mode
Purpose	Create & edit layouts	Test & preview the end-user experience
Canvas	Shows grid, snap guides, selection handles	Clean LCD rendering (no guides)
Click	Selects widget for editing	Simulates button press
Arrow keys	Nudge selected widget	Navigate pages / edit values
Toolbox	Visible (left panel)	Hidden (debug panel shows instead)
Properties	Visible (right panel)	Hidden (or shows register browser)
Who uses it	Engineer / Designer	Tester / End user